

LESSON 6-3 • SESSION 2 • PREREQUISITE PRACTICE A • SKILL 1 OF 2

Multiply Multi-Digit Numbers

Builds on Grade 4 • 4.NBT.B.5

TODAY'S LESSONYou will find $5^4 = 5 \times 5 \times 5 \times 5$. The products grow fast: 25, 125, 625.**YOU NEED TO**Multiply a 2- or 3-digit number by a 1-digit number, like 125×5 . **You will use it in Practice Form A, problems 3 and 4.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Break** the big number into hundreds, tens and ones.
- 2 Multiply** each part by the 1-digit number.
- 3 Add** the parts.

WATCH ME*the steps in action***Find 125×5 .****Step 1** $125 = 100 + 20 + 5$.**Step 2** $100 \times 5 = 500$, $20 \times 5 = 100$, $5 \times 5 = 25$.**Step 3** $500 + 100 + 25 = 625$.**Answer 625****TRY IT WITH HELP***fill in each step***1.** Find 27×3 .**Step 1** $27 = 20 + \underline{\hspace{2cm}}$ **Step 2** $20 \times 3 = \underline{\hspace{2cm}}$ and $7 \times 3 = \underline{\hspace{2cm}}$ **Step 3** $60 + 21 = \underline{\hspace{2cm}}$ **2.** Find 64×4 .**Step 1** $64 = 60 + \underline{\hspace{2cm}}$ **Step 2** $60 \times 4 = \underline{\hspace{2cm}}$ and $4 \times 4 = \underline{\hspace{2cm}}$ **Step 3** $\underline{\hspace{2cm}} + 16 = \underline{\hspace{2cm}}$ **YOUR TURN***use the steps on your own***3.** Multiply.

a. $25 \times 5 = \underline{\hspace{2cm}}$

b. $32 \times 2 = \underline{\hspace{2cm}}$

c. $49 \times 7 = \underline{\hspace{2cm}}$

d. $81 \times 3 = \underline{\hspace{2cm}}$

e. $36 \times 6 = \underline{\hspace{2cm}}$

f. $128 \times 2 = \underline{\hspace{2cm}}$

Extend Multiplying Patterns

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TODAY'S LESSON

You will complete a table of powers of 2. Each value is 2 times the one before.

YOU NEED TO

Continue a pattern that multiplies by the same number, like 1, 3, 9, 27. **You will use it in Practice Form A, problems 2 and 8.**

STEPS TO FOLLOW*use these same steps on every problem*

- 1** Find the **rule**: what is each number multiplied by?
- 2** **Multiply** the last number by the rule.
- 3** **Repeat** for the next blank.

WATCH ME*the steps in action*

Continue: 1, 2, 4, 8, __, __

Step 1 The rule is $\times 2$.

Step 2 $8 \times 2 = 16$.

Step 3 $16 \times 2 = 32$.

Answer 16, 32

TRY IT WITH HELP*fill in each step*

4. Continue: 1, 3, 9, __, __

Step 1 The rule is $\times$ _____

Step 2 $9 \times 3 =$ _____

Step 3 $27 \times 3 =$ _____

5. Continue: 1, 10, 100, __, __

Step 1 The rule is $\times$ _____

Step 2 $100 \times 10 =$ _____

Step 3 $1,000 \times 10 =$ _____

YOUR TURN*use the steps on your own*

6. Write the next two numbers.

a. 1, 5, 25, ... _____

b. 2, 4, 8, ... _____

c. 1, 4, 16, ... _____

d. 3, 6, 12, ... _____

READY CHECK*a problem like the one in today's lesson*

7. A table of powers of 2 shows 2, 4, 8, 16. What comes next?

A 18

B 24

C 32

D 64

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____